# Vishal Sharma


**Contact Information**

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Memberships: ACM, IEEE Senior, New York Academy of Sciences
•Website • Profile • Github • Kaggle • Quora


## Education & Training

**Postdoctoral Research Fellow** - Harvard Ophthalmology AI Lab — 2021 - 2022
Harvard Medical School/MEEI, Massachusetts, USA
*Mentor*: Mengyu Wang, PhD (Assistant Professor, Harvard Medical School/MEEI)

**Doctor of Philosophy**, **Computer Science** — 2016 - 2021 | 4.0
Utah State University, Utah, USA
*Dissertation*: Deep Learning Data and Indexes in a Database (over 1,500 downloads)
*Mentor*: Curtis Dyreson, PhD (Professor)

**Master of Science**, **Computer Science** — 2012 - 2014 | 3.91
Utah State University, Utah, USA
*Thesis*: MultiverseJava<temporal>: Programming Databases with Interesting Values
*Mentor*: Curtis Dyreson, PhD (Professor)

**Bachelors of Technology**, **Computer Science and Engineering** — 2006 - 2010 | 8.24
SRM Institute of Science and Technology, India (ranked 13 by the Ministry of Education, 2024)
First Class with Distinction (formerly SRM University)

## Academic Appointment

**Asst. Prof. TT, Big Data Science** - University of Nevada, Las Vegas — Feb 2024 - July 2026
- Part of campus-wide cluster hire. Developed four courses on Data Science for Healthcare track, part of a certificate program for Healthcare Information Technology
- Teaching: (i) Data Mining for Healthcare (rating 4.67/5) (ii) AI for Healthcare (4.35/5)

**Asst. Prof. TT, Computer Science** - New York Institute of Technology (NYIT) — Sept 2023 - Jan 2024
- Teaching: (i) Introduction to Data Mining (rating 4.8/5) (ii) Database Systems (rating 4.5/5)

**Research Associate/Scientist** - Cincinnati Children's — Oct 2022 - Dec 2022

## Teaching Experience

| | | | |
|---|---|---|---|
| **2025** | Instructor (4.67/5) | Ph.D. & Masters | Data Mining for Healthcare |
| **2025** | Instructor (4.35/5) | Ph.D. & Masters | Artificial Intelligence for Healthcare |
| **2024** | Instructor (4.41/5) | Ph.D. & Masters | Data Mining for Healthcare |
| **2023** | Instructor (4.5/5) | Ph.D. & Masters | Database Systems |
| **2023** | Instructor (4.8/5) | Masters & UG | Introduction to Data Mininig |

## Grants Preparation

**NIH R01** - Quantifying the impacts of social media on global mental health (under preparation)
**NSF CAREER** - Framework to Support The Process of Discovery (under preparation)
**NSF ART** - Virtual Reality (VR) Accelerated Ophthalmology (under preparation)

## Professional Experience

**Co Founder & CTO** - Vaision, USA — 2023 - 2024
- Vaision has been concluded

**Research Intern** - Intel, USA — Summer 2019
- *Mentor*: Don Kent (Senior Manager Data Science)

**Data Scientist Intern** - IM Flash (Intel-Micron), USA — Summer 2018
- *Mentor*: Pradeep Ramachandran (Senior Member of Technical Staff)

**Senior Software Engineer** - InMoment, USA — Mar 2015 - April 2016
**Software Engineer** - McAfee, USA — Oct 2014 - Mar 2015
**Asst. System Engineer/Role: Software Engineer** - Tata Consultancy Services, INDIA — Sept 2010 - July 2012

## Publications/Abstracts/Posters/Pre-print/Github

[27] **An Approach for Cross Platform Workload for Autonomous Database Systems**
Vishal Sharma
*World Academy of Science, Engineering and Technology (WASET)*, 2026

[26] **A Framework to Unify Lost Knowledge**
Vishal Sharma
*World Academy of Science, Engineering and Technology (WASET)*, 2026

[25] **The Metadata Impedance Mismatch between Databases and Programming Languages**
Vishal Sharma, Curtis Dyreson
$34^{th}$ *Conference on Information and Knowledge Management*, *CIKM*, 2025

[24] **Reproducibility Report for ACM SIGMOD 2024 Paper: "Temporal JSON Keyword Search"**
Chenhao Ma, Junchang Wang, Curtis Dyreson, Amani Shatwani, Sourav S Bhowmick, Vishal Sharma
$50^{th}$ *ACM International Conference on Management of Data*, Reproducibility Reports of the 2024 International Conference on Management of Data, 2024

[23] **Temporal JSON Keyword Search**
Curtis Dyreson, Amani Shatwani, Sourav S Bhowmick, Vishal Sharma
⋆ *Awards: Artifact Received, Artifact Evaluated and Results Reproduced*
$50^{th}$ *ACM International Conference on Management of Data*, *SIGMOD*, 2024

[22] **Artifact-Tolerant Clustering-Guided Contrastive Embedding Learning for Ophthalmic Images**
Min Shi, Anagha Lokhande, Mojtaba S Fazli, Vishal Sharma, Yu Tian, Yan Luo, Louis R Pasquale, Tobias Elze, Michael V Boland, Nazlee Zebardast, David S Friedman, Lucy Q Shen, Mengyu Wang
*IEEE Journal of Biomedical and Health Informatics*, 2023 (IF: 7.1)

[21] **Indexer++: Workload-Aware Online Index Tuning with Transformers and Reinforcement Learning**
Vishal Sharma, Curtis Dyreson
$37^{th}$ *ACM SIGAPP Symposium on Applied Computing*, *SAC*, 2022 (AR: 22%)

[20] **PyVisualFields: A Python Package for Visual Field Analysis**
Mohammad Eslami, Saber Kazeminasab, Vishal Sharma, Yangjiani Li, Mojtaba Fazli, Mengyu Wang, Nazlee Zebardast, and Tobias Elze
*Translational Vision Science & Technology*, TVST, ARVO, 2022 (IF: 3.28)

[19] **Mantis: Multiple Type and Attribute Index Selection using Deep Reinforcement Learning**
Vishal Sharma, Curtis Dyreson, Nicholas Flann
$25^{th}$ *ACM International Database Engineering & Applications Symposium*, IDEAS 2021 (AR: 28%)

[18] **Popularity vs Quality: Analyzing and Predicting the Success of Highly Rated Crowdfunded Projects on Amazon**
Vishal Sharma, Kyumin Lee, Curtis Dyreson
*Springer Nature Computing*, 2021 (IF: 3.7)

[17] **Automating and Analyzing Whole-Farm Carbon Models**
Aditi Maheshwari, Curtis Dyreson, Jennifer Reeve, Vishal Sharma, Anthony Whaley
$7^{th}$ *IEEE International Conference on Data Science and Analytics*, DSAA 2020 (AR: 27%)

[16] **Covid-19 Screening Using Residual Attention Network an Artificial Intelligence Approach**
Vishal Sharma, Curtis Dyreson
$19^{th}$ *IEEE International Conference on Machine Learning and Applications*, ICMLA 2020 (AR: 25%)

[15] **LinkSocial: Linking User Profiles Across Multiple Social Media Platforms**
Vishal Sharma, Curtis Dyreson
$8^{th}$ *IEEE International Conference on Big Knowledge, ICBK (in conjunction with ICDM)* 2018 (AR: 27%)

[14] **Predicting Highly Rated Crowdfunded Products**
Vishal Sharma, Kyumin Lee
$10^{th}$ *IEEE/ACM Advances in Social Networks Analysis and Mining*, ASONAM 2018 (AR: 16%)

[13] **Recommending Prime Spots of a Destination and Time to Visit from Geo-tagged Social Data**
Vishal Sharma, Kyumin Lee, Jinwook Chung
$10^{th}$ *IEEE International Conference on Collaborative Computing: Networking, Applications and Worksharing*, CollaborateCom 2014 (AR: 28%)

[12] **Supporting data aspects in pig latin**
Curtis Dyreson, Omar U. Florez, Akshay Thakre, Vishal Sharma
$12^{th}$ *ACM Aspect-oriented Software Development*, AOSD 2013 (AR: 25%)

[11] **The Multiverse Programming Paradigm: Programming with Values Annotated with Metadata**,
Vishal Sharma,
Graduate Research Symposium. Paper 87. https://digitalcommons.usu.edu/grs/87, (2014).

[10] **Role of generative artificial intelligence (AI) in generating course design and curriculum in dental public health**
Jyoti Mago, Vishal Sharma, Manoj Sharma
*American Public Health Association*, APHA, 2024

[9] **Role of generative artificial intelligence (AI) in oral health**
Jyoti Mago, Vishal Sharma, Manoj Sharma
*American Public Health Association*, APHA, 2024

[8] **Entity and Relation Identification in Unstructured Endoscopy Reports**
Nicholas Denson, Judith W Dexheimer, Julia Smith, Ryan Murphy, Vishal Sharma, Liu Xiaoxuan, Surya Prasath, Yizhao Ni, James Reigle, and Jasbir Dhaliwal
⋆ *Poster of Distinction*, *Digestive Disease Week*, $DDW$, $2023(IF: 3.49)$

[7] **A Deep Autoencoder Model to Denoise Visual Fields in Glaucoma**
Vishal Sharma, Lucy Q Shen, Louis Pasquale, Tobias Elze, Michael V Boland, Sarah R Wellik, Gustavo De Moraes, Jonathan S Myers, Siamak Yousefi, Mengyu Wang
⋆ *Association for Research in Vision and Ophthalmology (ARVO) foundation travel grant, Denver, CO, 2022 (IF: 2.39)*

[6] **A Python Collection of Tools for Analyzing Visual Fields**
Saber Kazeminasab, Mohammad Eslami, Yangjiani Li, Mojtaba Fazli, Vishal Sharma, Mengyu Wang, Nazlee Zebardast, Tobias Elze
*Association for Research in Vision and Ophthalmology*, ARVO 2022 (IF: 2.39)

[5] **Evaluation of Deep Learning Visual Field Prediction Models for Clinical Relevance**
Mohammad Eslami, Miao Zhang, Julia Kim, Dolly Chang, Yangjiani Li, Saber Kazeminasab, Mojtaba Fazli, Vishal Sharma, Michael Boland, Nazlee Zebardast, Mengyu Wang, Tobias Elze
*Association for Research in Vision and Ophthalmology*, ARVO 2022 (IF: 2.39)

[4] **Glaucomatous Progressive Retinal Nerve Fiber Layer Thinning and Its Association With Patient Race**
Qingying Jin, Omar Halawa, Yangjiani Li, Mohammad Eslami, Saber Kazeminasab, Mojtaba Fazli, Vishal Sharma, Nazlee Zebardast, Mengyu Wang, Tobias Elze
*Association for Research in Vision and Ophthalmology*, ARVO 2022 (IF: 2.39)

[3] **The Impact of Race on the Relationship Between Cup-To-Disc Ratio and Glaucomatous VF Los**
Pingping Zhao, Yangjiani Li, Mohammad Eslami, Saber Kazeminasab, Mojtaba Fazli, Vishal Sharma, Omar Halawa, Nazlee Zebardast, Mengyu Wang, Tobias Elze
*Association for Research in Vision and Ophthalmology*, ARVO 2022 (IF: 2.39)

[2] **Speaker Diarization: Using Recurrent Neural Networks**
Vishal Sharma, Zekun Zhang, Zachary Neubert, Curtis Dyreson
*arXiv:2006.05596, preprint*, Github 2020

[1] **Multi Class Audio Classification Using Multi Layer Perceptron and Convolution Neural Network**
Vishal Sharma
https://doi.org/10.5281/zenodo.3988690, Github , 2020

## Invited Research Presentations

| | |
|---|---|
| **2025** | ACM International Conference on Information and Knowledge Management, CIKM, 2025, Seoul, Korea |
| **2022** | ACM SIGAPP Symposium On Applied Computing (SAC), Czech Republic (online) |
| **2021** | ACM International Database Engineering & Applications Symposium (IDEAS), Montreal (online) |
| **2020** | IEEE International Conference on Machine Learning and Applications (ICMLA), Miami (online) |
| **2018** | IEEE International Conference on Big Knowledge (ICBK/ICDM), Singapore |
| **2018** | IEEE International Conference on Advances in Social Networks Analysis & Mining (ASONAM), Barcelona |

## Academic Service & Leadership

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|---|---|---|
| **2025** | Reviewer | Int'l Journal of Software Engineering and Knowledge Engineering |
| **2025** | Technical PC | $11^{th}$ International Conference on Human and Social Analytics |
| **2024** | Technical PC | $10^{th}$ International Conference on Human and Social Analytics |
| **2023** | Technical PC | $9^{th}$ International Conference on Human and Social Analytics |
| **2022** | Reviewer | Review of Hypermedia and Multimedia (NRHM) |
| **2022** | Technical PC | $8^{th}$ International Conference on Human and Social Analytics |
| **2021** | Program Committee | IEEE BIBM Artificial Intelligence Techniques for BioMedicine and Health |
| **2021** | Technical PC | $7^{th}$ International Conference on Human and Social Analytics |
| **2020** | Program Committee | IJCAI Artificial Intelligence in Affective Computing (AffComp) |
| **2020** | Program Committee | IEEE BIBM Artificial Intelligence Techniques for BioMedicine and Health |
| **2020** | Program Committee | IEEE BIBM Artificial Intelligence & Big Data vs Pandemics |
| **2020** | Technical PC | $6^{th}$ International Conference on Human and Social Analytics |
| **2019** | Technical PC | $5^{th}$ International Conference on Human and Social Analytics |
| **2018** | Session Chair | ACM/IEEE ASONAM |
| **2014** | Student Volunteer | ACM SIGMOD |

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